

Automatic Rigidity for Collatz-Invariant 7-Automatic Sequences

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Lean 4 formally verified version 1.1 (the ARC7 main proof does not use Cobham's theorem)

September 18, 2026

Abstract

Let $a = (a_n)_{n \geq 0}$ be a sequence over a finite alphabet. We prove that if a is 7-automatic and satisfies

$$a_{2n} = a_n, \quad a_{2n+1} = a_{3n+2} \quad (n \geq 0),$$

then a is constant on the positive integers, while a_0 remains genuinely free. Equivalently, every finite-alphabet 7-automatic coloring invariant under the shortcut Collatz map is trivial on $\mathbb{N}_{>0}$.

The base-7 argument follows the local/dyadic route developed for ARC5, but a new arithmetic obstruction appears in the finite-state pumping step: $7^L \equiv 3 \pmod{4}$ when L is odd. The obstruction is removed by pumping a doubled loop. If the original loop has length L , then the doubled block has length $2L$ and $7^{2L} \equiv 1 \pmod{8}$. This produces an exact dyadic factorization of the numerical pumping increments. The resulting pumped family fills every sufficiently deep dyadic child of an appropriate parent cylinder. Hence an infinite 7-automatic Boolean support cannot be dyadically nowhere thick.

Applying this topological pumping theorem to the adjacent-change set $D_a = \{n : a_n \neq a_{n+1}\}$ gives finiteness of D_a : shortcut invariance makes D_a dyadically nowhere thick, while finite 7-kernel of a implies finite 7-kernel of its characteristic sequence, including the carry case at the boundary of a base-7 residue block. Therefore a is eventually constant. Finally, the identity $a_{2n} = a_n$ propagates the tail value back to every positive integer.

The complete proof chain was formalized in Lean 4 / Mathlib, culminating in `arc7_final`. A separate formal sharpness theorem proves that for arbitrary x, y , the sequence $a_0 = x$ and $a_n = y$ for $n \geq 1$ is 7-automatic and shortcut-Collatz invariant. The active ARC7 files contain no `sorry`, no `admit`, no custom axiom declaration, and the ARC7 main proof does not invoke the Cobham principle.

Keywords: Collatz conjecture; automatic sequence; finite automata; 7-kernel; invariant coloring; pumping; 2-adic topology; dyadic residue filling; Lean 4.

2020 Mathematics Subject Classification: Primary 11B85; Secondary 11B83, 68Q45.

Author's note: how this study arose

This work is the next stage of an exploration of automatic rigidity for Collatz-invariant sequences in bases 2, 3, 5, and 7. The base-3 proof admits a short cross-base kernel route, whereas ARC5 forced a different mechanism built from Collatz local synchronization, dyadic thinness, and finite-state pumping. ARC7 was undertaken as a stress test of that second mechanism.

The first apparent obstacle was genuinely arithmetic. In base 5, a loop of length L produces the factor $5^L \equiv 1 \pmod{4}$ automatically. In base 7, the congruence oscillates with the parity of L . The decisive idea was to replace a pump loop by its doubled copy, so that the effective base factor becomes $7^{2L} \equiv 1 \pmod{8}$. The formal development was then organized into small stages isolating doubled-loop pumping, exact dyadic factorization, residue permutation, child filling, change-set automaticity, and final rigidity.

I am not a professionally trained mathematician. The project has been developed through sustained discussion with a contemporary conversational AI, with repeated adversarial checking and Lean compilation used to expose and repair gaps. The human-readable proof below is intentionally compressed relative to the formal development; the Lean files expand the finite-state, arithmetic, dyadic-cylinder, carry, and interface arguments into machine-checked steps.

1 Introduction

Define the shortcut Collatz map $T : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ by

$$T(2n) = n, \quad T(2n+1) = 3n+2 \quad (n \geq 0). \quad (1)$$

On positive integers this is the usual $3x+1$ map with one division by 2 incorporated into the odd step. For background on the $3x+1$ problem, see Lagarias [10].

Let Δ be a finite alphabet. A coloring $a : \mathbb{N}_0 \rightarrow \Delta$ is T -invariant if

$$a_n = a_{T(n)} \quad (n \geq 0).$$

By (1), this is equivalent to

$$a_{2n} = a_n, \quad a_{2n+1} = a_{3n+2} \quad (n \geq 0). \quad (2)$$

Finite-state descriptions of Collatz-related sets and relations have been studied from several directions. Shallit and Wilson [11] and Sterin [12] studied automata associated with Collatz predecessor or ancestor structures. Sterin and Woods [13] exhibited a base-conversion phenomenon inside the Collatz process. Caucal and Rispal [3] studied transducer descriptions for powers of Collatz-type maps. Related questions about recognizability and affine closure appear in [2, 7, 8, 9].

The present problem is different. We do not ask an automaton to compute T , nor to recognize a bounded-depth orbit family. We ask whether a finite-valued coloring that is constant along every edge of the full functional graph of T can itself be 7-automatic without being trivial.

Theorem 1.1 (Automatic rigidity in base 7; ARC7). *Let Δ be a finite alphabet and let $a = (a_n)_{n \geq 0} \in \Delta^{\mathbb{N}_0}$ be a 7-automatic sequence. Assume (2) for every $n \geq 0$. Then there exists $c \in \Delta$ such that*

$$a_n = c \quad (n \geq 1).$$

Theorem 1.1 makes no assumption about the truth of the Collatz conjecture. Its proof separates into two forces:

1. *Collatz local synchronization*: shortcut invariance creates monochromatic local blocks inside arbitrarily prescribed dyadic neighborhoods;
2. *base-7 finite-state pumping*: an infinite 7-automatic Boolean support cannot remain dyadically nowhere thick, because a doubled pump loop produces a family that fills all sufficiently deep dyadic children of a parent cylinder.

Their combination makes the adjacent-change set finite. Eventual constancy then follows, and doubling invariance upgrades eventual constancy to constancy on all positive inputs.

Remark 1.2 (A concrete local-synchronization example). The local synchronization mechanism is already visible in a small exact identity. For every $m \geq 0$,

$$\begin{aligned} 8m+4 &\mapsto 4m+2 \mapsto 2m+1 \mapsto 3m+2, \\ 8m+5 &\mapsto 12m+8 \mapsto 6m+4 \mapsto 3m+2. \end{aligned}$$

Hence

$$T^3(8m+4) = T^3(8m+5) = 3m+2.$$

Any T -invariant coloring therefore satisfies $a_{8m+4} = a_{8m+5}$ on the entire dyadic class $n \equiv 4 \pmod{8}$. This toy identity is not the general local-density theorem used in the proof, but it illustrates the same principle: after refining to a suitable dyadic child, two nearby inputs can be forced to follow compatible branch patterns and coalesce uniformly.

2 Automatic sequences, kernels, and the base-7 obstruction

For an integer $k \geq 2$ and a sequence $u = (u_n)_{n \geq 0}$, define the k -kernel

$$\mathcal{K}_k(u) := \{(u_{k^e n + r})_{n \geq 0} : e \geq 0, 0 \leq r < k^e\}.$$

A finite-alphabet sequence is k -automatic if and only if its k -kernel is finite [1]. The Lean development uses this finite-kernel definition internally and also proves the bridge to finite most-significant-digit DFAO presentations. In particular, for base 7 the final theorem is exported to the usual canonical MSD convention in which zero is represented by the one-digit word [0].

The base-3 ARC route exploits a special interaction between the coefficients 2 and 3 in (2): binary decimation of the 3-kernel can be kept inside a finite 3-kernel hierarchy, leading to a Cobham argument. The analogous direct closure is not available in base 7. If

$$u_n = a_{7^j n + r},$$

then binary decimation introduces a coefficient $2 \cdot 7^j$, and use of the odd branch can introduce $3 \cdot 7^j$. In general these are not powers of 7. Thus the obstruction is structural, not a missing simplification.

The ARC7 proof therefore follows the local/dyadic route. The base-independent Collatz part is inherited from the local-density development: for every finite collection of offsets and every prescribed dyadic parent class, one can refine to a positive child class on which all corresponding color equalities hold simultaneously. For the final argument only the offset 1 is required: inside every dyadic parent one can find a deeper positive child on which

$$a_n = a_{n+1}$$

holds identically.

3 Doubled-loop pumping arithmetic

The genuinely new base-7 issue enters in the automaton pumping argument. A single loop of length L naturally produces powers 7^L . Unlike base 5, one does not always have $7^L \equiv 1 \pmod{4}$; rather

$$7^L \equiv 3 \pmod{4} \quad (L \text{ odd}), \quad 7^L \equiv 1 \pmod{4} \quad (L \text{ even}).$$

The remedy is to replace the loop block b by the doubled block bb . Its effective length is $2L$, and

$$7^{2L} = (7^2)^L \equiv 1 \pmod{8}. \tag{3}$$

Let

$$G_q(k) := 1 + q + \cdots + q^{k-1}, \quad G_q(0) = 0.$$

The arithmetic model used in the formal proof is

$$N_k = N_0 + d G_{7^{2L}}(k). \tag{4}$$

Here N_0 is the unpumped numerical value, $d > 0$ is the first numerical increment, and the factor 7^{2L} comes from the doubled loop.

Lemma 3.1 (Exact dyadic displacement). *Assume*

$$d = 2^s(2v + 1).$$

For every $t, u, k \geq 0$ there exists $w \geq 0$ such that

$$N_{k+2^t(2u+1)} = N_k + 2^{s+t}(2w + 1).$$

Proof sketch. Split the geometric sum at k . Equation (3) implies that all powers of 7^{2L} are odd and, more strongly, congruent to 1 modulo 4. The geometric sum of $2^t(2u + 1)$ such terms therefore has exact 2-adic factor 2^t . Multiplying by $d = 2^s(2v + 1)$ gives the displayed factorization. This is the content of the Lean theorem `arc7_stage1_even_loop_arithmetic_core`.

The automaton side supplies the parameters in Lemma 3.1. A sufficiently long accepted word admits a standard pumping decomposition $x = abc$ with $b \neq \emptyset$. If reading b returns the DFA state to itself, then so does reading bb ; hence

$$a(bb)^k c$$

is accepted for every $k \geq 0$. The formal Stage 2 records this doubled-loop acceptance explicitly rather than merely invoking an abstract pumping lemma.

For canonical positive base-7 words, the numerical value of the pumped word is then identified exactly with the family (4). The first increment is proved strictly positive, and factored in the form

$$d = 2^s(2v + 1). \tag{5}$$

This produces the Stage-5 handoff: a support-preserving pumped numerical family together with exact dyadic displacement for every odd-times-power-of-two change in pump count.

4 Dyadic residue permutation and child filling

Normalize the increment in (5) by removing the factor 2^s . For a target depth M , consider the first 2^M pump indices. The formal residue map is represented by

$$k \mapsto N'_k \pmod{2^M}, \quad 0 \leq k < 2^M,$$

where N'_k is the pumped family with odd initial increment $2v + 1$.

Proposition 4.1 (Dyadic residue permutation). *For every $M \geq 0$, the first 2^M normalized pumped values hit every residue modulo 2^M exactly once.*

Proof sketch. Suppose two indices $i < j < 2^M$ give the same residue. Write $j - i = 2^t(2u + 1)$. Lemma 3.1 gives an exact difference $2^t(2w + 1)$ after normalization. Since $j - i < 2^M$, one has $t < M$, so that difference cannot vanish modulo 2^M . Hence the residue map is injective. Its domain and codomain both have 2^M elements, so it is bijective. This is formalized as `arc7_stage6_dyadic_residue_permutation`.

Restoring the factor 2^s shows that the original pumped family fills all children of its parent dyadic cylinder.

Proposition 4.2 (Dyadic child filling). *Let N_k be a pumped family with increment $d = 2^s(2v + 1)$. For every depth increment M , every dyadic child modulo 2^{s+M} of the parent class of N_0 modulo 2^s contains some pumped value N_k .*

This is stronger than mere density along a subsequence: the first 2^M normalized pump indices already provide a complete transversal of the deeper children.

Definition 4.3 (Dyadically nowhere thick). A set $S \subseteq \mathbb{N}_0$ is called dyadically nowhere thick if, inside every dyadic parent residue class, there exists a deeper positive dyadic child residue class contained in the parent and disjoint from S .

The proof uses this explicit combinatorial definition. No identification with every possible standard topological formulation in $\mathbb{Z}_2$ is needed.

Proposition 4.4 (Pumped support defeats thinness). *If all values of a pumped family satisfying Proposition 4.2 lie in S , then S is not dyadically nowhere thick.*

Indeed, apply Definition 4.3 to the parent class of the pumped family. The supposedly empty deeper child is one of the children hit by Proposition 4.2, a contradiction.

5 Topological pumping for 7-automatic supports

Let $s : \mathbb{N}_0 \rightarrow \{0, 1\}$ be generated by a finite MSD-DFAO, and let $S = \text{supp}(s)$. If S is infinite, then it contains arbitrarily large integers and therefore canonical base-7 support words of arbitrarily large length. In particular there is a positive canonical support word whose length is at least the number of automaton states.

Applying the pumping lemma to such a word and then Stages 3–8 yields a support-preserving doubled-loop family of the type considered above. Proposition 4.4 then gives the central compactness statement.

Theorem 5.1 (Base-7 topological pumping). *Let $s : \mathbb{N}_0 \rightarrow \{0, 1\}$ be 7-automatic. If $\text{supp}(s)$ is dyadically nowhere thick, then $\text{supp}(s)$ is finite.*

The Lean development first proves the theorem for an explicit finite MSD-DFAO presentation (`arc7_stage9_msd_topological_pumping`), then transports it through the finite-kernel/MSD bridge to obtain `arc7_stage10_kernel_topological_pumping`.

Theorem 5.1 does not use Cobham’s theorem. Its compactness mechanism is internal: finite-state pumping collides with exact dyadic residue filling.

6 Automaticity of the adjacent-change indicator

Define the adjacent-change set

$$D_a := \{n \geq 0 : a_n \neq a_{n+1}\} \quad (6)$$

and its Boolean characteristic sequence χ_a .

The only subtlety in proving 7-automaticity of χ_a is the carry at the right edge of a base-7 residue block. Consider a 7-kernel residual

$$n \longmapsto \chi_a(7^e n + r), \quad 0 \leq r < 7^e.$$

Its two compared a -values are

$$a_{7^e n + r} \quad \text{and} \quad a_{7^e n + r + 1}.$$

If $r + 1 < 7^e$, both are ordinary 7-kernel states. If $r + 1 = 7^e$, then

$$7^e n + r + 1 = 7^e(n + 1), \quad (7)$$

so the second term is the one-step successor of the zero-residue kernel state $n \mapsto a_{7^e n}$.

Thus every kernel residual of χ_a is a disagreement indicator comparing one ordinary kernel state with either another ordinary kernel state or a one-step successor of one. A finite kernel has only finitely many such states and successors, hence only finitely many such comparisons.

Proposition 6.1 (Automaticity of the change indicator). *If a has finite 7-kernel, then χ_a has finite 7-kernel. Equivalently, the adjacent-change indicator of a 7-automatic sequence is 7-automatic.*

This is the content of `arc7_stage11_change_automaticity`. Equation (7) is the formal carry branch checked explicitly in that proof.

7 Collatz invariance makes the change set finite

We now use the base-independent local-synchronization theorem inherited from the Collatz part of the ARC5 development. In the form needed here, shortcut invariance alone implies the following.

Lemma 7.1 (Local adjacent synchronization). *Let a satisfy (2). Inside every prescribed dyadic parent class there exists a deeper positive dyadic child class on which*

$$a_n = a_{n+1}$$

holds for every n in that child.

The formal proof invokes the finite simultaneous local-density theorem with block bound $H = 1$. No base-7 automaticity is used in Lemma 7.1.

By definition of D_a , Lemma 7.1 says precisely that every dyadic parent contains a deeper positive child disjoint from D_a .

Corollary 7.2. *If a is shortcut-Collatz invariant, then D_a is dyadically nowhere thick.*

Combining Corollary 7.2, Proposition 6.1, and Theorem 5.1 gives the decisive finite-change conclusion.

Theorem 7.3 (Finite adjacent-change set). *If a is 7-automatic and satisfies (2), then D_a is finite.*

This chain is packaged in Lean as `arc7_stage12_change_set_finite_handoff`. An important point is that the completed ARC7 proof does not need an intermediate theorem asserting dyadic thinness for every pair of same-level kernel states. The final proof acts directly on the adjacent-change set.

8 Eventual constancy and positive rigidity

Finiteness of D_a immediately implies eventual constancy. Choose N strictly larger than every element of D_a . Then

$$a_n = a_{n+1} \quad (n \geq N),$$

so induction gives a single value c with

$$a_n = c \quad (n \geq N). \tag{8}$$

This is formalized in `arc7_stage13_eventual_constancy`.

It remains to transport the tail value back to all positive indices. The even branch of (2) gives

$$a_{2n} = a_n,$$

and therefore, by induction,

$$a_{2^k n} = a_n \quad (k, n \geq 0). \tag{9}$$

For any $n > 0$, choose k so large that $2^k n \geq N$. Then (8) and (9) imply

$$a_n = a_{2^k n} = c.$$

This proves Theorem 1.1; the corresponding Lean handoff is `arc7_stage14_positive_rigidity`, and the packaged endpoint is `arc7_final`.

9 Sharpness at zero and automatic invariant sets

The exclusion of zero in Theorem 1.1 is not a weakness of the proof; it is sharp.

Proposition 9.1 (The zero value is genuinely free). *Let x, y be arbitrary values. Define*

$$b_0 = x, \quad b_n = y \quad (n \geq 1).$$

Then b is 7-automatic and satisfies

$$b_{2n} = b_n, \quad b_{2n+1} = b_{3n+2}$$

for every $n \geq 0$.

Proof. The two branch identities are immediate because every positive index has value y , while the even branch maps zero to zero. The 7-kernel contains at most two sequences: b itself and the constant sequence y . Hence the kernel is finite. The Lean theorem `arc7_zero_value_is_genuinely_free` proves the statement in the usual canonical MSD automaticity interface.

A set-theoretic version follows by applying Theorem 1.1 to a Boolean characteristic sequence. Thus a 7-automatic subset of $\mathbb{N}_0$ invariant under the shortcut Collatz edges is, on positive integers, either empty or all of $\mathbb{N}_{>0}$; membership of zero remains independent. This is a rigidity statement for automatic invariant sets, not a resolution of the Collatz conjecture.

10 Relation to previous work and scope of the result

The literature contains many automata-theoretic and logical descriptions of Collatz-related structures, but they address different questions: computation of iterations, predecessor sets, base conversion, transducers, affine recognizable sets, or definability. The present theorem concerns a one-variable finite coloring that is invariant along every shortcut-Collatz edge and is automatic in base 7.

The proof mechanism should therefore be distinguished from two nearby ideas. First, the result does not assert that the Collatz map itself is automatic in base 7. Second, it does not use a general theorem saying that every thin automatic set is finite in every numeration system; the proof establishes the exact base-7 support statement needed here by doubled-loop pumping and dyadic residue filling.

Targeted literature searches during the development did not identify a theorem stated in the same form as Theorem 1.1 or Theorem 5.1. This is a provisional literature conclusion only. No claim of exhaustive priority is made here; an equivalent consequence of a more general theorem under different terminology may exist.

The doubled-loop observation also suggests a broader program. Every odd integer b satisfies $b^2 \equiv 1 \pmod{8}$, so the first arithmetic obstruction encountered at base 7 disappears after loop doubling for any odd base. However, the present paper proves only the base-7 theorem. A general odd-base ARC theorem is left for future work and is not inferred merely from the base-7 formalization.

11 Comparison with ARC2, ARC3, and ARC5

The ARC cases reach the same positive-domain rigidity conclusion through different bridges.

	ARC2	ARC3	ARC5	ARC7
Starting structure	2-automatic	3-automatic	5-automatic	7-automatic
Main bridge	suitable 3-kernel family inside finite 2-kernel	direct binary-decimation kernel transfer	local synchronization + dyadic thinness + pumping	doubled-loop pumping + dyadic filling + change set
Global compactness	Cobham	Cobham	finite change set	finite change set
Key arithmetic	powers of 2 modulo powers of 3	coefficient matches the kernel base	$5^L \equiv 1 \pmod{4}$	$7^{2L} \equiv 1 \pmod{8}$
Final propagation	Collatz identities	Collatz identities	doubling invariance	doubling invariance
Cobham in main proof	used	used	not used	not used

The ARC7 argument is therefore not merely a textual replacement of 5 by 7. The parity oscillation of 7^L forces a structural change in the pumping unit, and the doubled-loop construction is the new bridge that restores exact dyadic control.

12 Formal verification in Lean 4

The ARC7 proof chain was developed in Lean 4 / Mathlib. The final entry point is

`arc7_final,`

whose hypotheses are shortcut invariance and finite base-7 kernel, and whose conclusion is existence of one value on every positive natural input. The standard paper-facing automaticity interface is

`arc7_main_zero_digit_standard_msd_form.`

The project was built stage by stage: each new stage was first checked as a standalone Lean file, then imported into the project root and followed by a full `lake build`. The completed project build includes Stages 1–14, the final theorem, the automaticity audit, and the sharpness file `ARC7Sharpness_v3.lean`.

Formal theorem	Main verified content
<code>arc7_stage1_even_loop_arithmetic_core</code>	Exact dyadic displacement of the doubled-loop numerical family.
<code>arc7_stage2_double_loop_pumping_handoff</code>	Extraction and arbitrary pumping of a nonempty doubled automaton loop.
<code>arc7_stage3_base7_numeric_pumping_handoff</code>	Exact identification of doubled-pumped base-7 word values with the arithmetic family.
<code>arc7_stage4_positive_numeric_pumping_handoff</code>	Strict positivity of the numerical increment for positive canonical words.
<code>arc7_stage5_exact_dyadic_pumping_handoff</code>	Positive increment, exact $2^s \times \text{odd}$ factorization, support preservation, and displacement package.
<code>arc7_stage6_dyadic_residue_permutation</code>	Bijectivity of the normalized pumped residue map modulo 2^M .
<code>arc7_stage7_dyadic_child_filling</code>	Every deeper dyadic child of the parent cylinder is hit.
<code>arc7_stage8_long_word_collision</code>	One sufficiently long positive canonical support word contradicts dyadic nowhere-thickness.
<code>arc7_stage9_msd_topological_pumping</code>	MSD-DFAO topological pumping theorem.
<code>arc7_stage10_kernel_topological_pumping</code>	Finite-kernel version: nowhere-thick 7-automatic Boolean support is finite.
<code>arc7_stage11_change_automaticity</code>	Adjacent-change indicator is 7-automatic; carry boundary handled explicitly.
<code>arc7_stage12_change_set_finite_handoff</code>	Change set is automatic, dyadically nowhere thick, and finite.
<code>arc7_stage13_eventual_constancy</code>	Finite change set implies an eventually constant tail.
<code>arc7_stage14_positive_rigidity</code>	Doubling invariance propagates the tail value to all positive integers.
<code>arc7_final</code>	Packaged ARC7 rigidity theorem.
<code>arc7_kernel_iff_zero_digit_standard_msd</code>	Equivalence with the usual canonical MSD automaticity interface.
<code>arc7_zero_value_is_genuinely_free</code>	Formal sharpness: a_0 can be chosen independently.

Cobham and external assumptions

Unlike ARC2 and ARC3, the ARC7 main chain does not invoke the Cobham principle. The finite-kernel/MSD conversion used by the topological pumping stage is proved inside the ARC formal development. Searches of the active ARC7 source files found no occurrence of `ARCCobhamPrinciple` in the proof chain.

Axiom auditing

The principal ARC7 files contain explicit `#print axioms` commands. The final rigidity, standard-automaticity audit, and sharpness theorems report only the standard dependencies `propext`, `Classical.choice`, and `Quot.sound` (with some elementary theorems using a subset of these). No `sorryAx` appears in the successful final builds. The active ARC7 source files used for the project contain no `sorry`, no `admit`, and no custom `axiom` declaration.

This verifies formal dependency closure of the stated Lean development. In addition, the theorem statements, shortcut-map definition, automaticity interfaces, carry boundary, change-set construction, final positive-rigidity step, and sharpness statement were separately compared against the manuscript during a post-formalization semantic audit. Within the formal and semantic audits performed for this version, no concrete counterexample or fatal logical inconsistency has been identified. This is deliberately weaker than a claim of mathematical infallibility: formal verification does not, by itself, replace independent mathematical peer review or continued literature auditing.

Disclosure of generative-AI assistance

Generative AI (OpenAI ChatGPT, GPT-5.6 Sol) was used in preparing this manuscript for literature-search assistance, theorem formulation, adversarial checking of proof steps, editorial

drafting, assistance with Lean 4 proof-code development, diagnosis of compilation errors, and post-formalization auditing. The AI system is not an author and is not used as a bibliographic source. Responsibility for the mathematical claims, references, formalization, and final manuscript rests with the human author.

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